

Permutations & Combinations

ANSWER KEY



The factorial function and counting arrangements

- 1 In how many ways can 6 different books be arranged in a row on a shelf?
 $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ ways.
 - 2 In how many ways can 5 different keys be arranged in a row on a keyring holder?
 $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ways.
 - 3 In how many ways can 7 different trophies be arranged on a shelf?
 $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$ ways.
 - 4 In how many ways can 4 different paintings be hung in a row on a wall?
 $4! = 4 \times 3 \times 2 \times 1 = 24$ ways.
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Permutations of a subset of items

- 5 In how many ways can 4 runners be chosen from a squad of 9 and assigned to first, second, third, and fourth positions in a relay?
This is a permutation: ${}^9P_4 = \frac{9!}{(9-4)!} = \frac{9!}{5!} = 9 \times 8 \times 7 \times 6 = 3024$.
 - 6 In how many ways can 3 medals (gold, silver, bronze) be awarded to 3 of 8 competitors in a race?
This is a permutation: ${}^8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$.
 - 7 In how many ways can 5 books be chosen from 12 and arranged in order on a shelf?
This is a permutation: ${}^{12}P_5 = \frac{12!}{(12-5)!} = \frac{12!}{7!} = 12 \times 11 \times 10 \times 9 \times 8 = 95040$.
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Combinations

- 8 A committee of 4 people is to be chosen from a group of 10. In how many ways can this be done?
This is a combination (order doesn't matter): ${}^{10}C_4 = \frac{10!}{4!6!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = \frac{5040}{24} = 210$.
- 9 A hand of 5 cards is dealt from a standard deck of 52 cards. In how many ways can this be done?
 ${}^{52}C_5 = \frac{52!}{5!47!} = 2598960$.
- 10 A pizza shop offers 8 toppings. In how many ways can a customer choose 3 toppings for a pizza?
 ${}^8C_3 = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = \frac{336}{6} = 56$.
- 11 A student must answer 5 out of 7 questions on an exam. In how many ways can the student choose which questions to answer?
 ${}^7C_5 = \frac{7!}{5!2!} = \frac{7 \times 6}{2 \times 1} = 21$.

Arrangements with repeated items

- 12** In how many distinct ways can the letters of the word "BANANA" be arranged?
"BANANA" has 6 letters, with A repeated 3 times and N repeated 2 times. Number of arrangements
$$= \frac{6!}{3! 2!} = \frac{720}{12} = 60.$$
- 13** In how many distinct ways can the letters of the word "MISSISSIPPI" be arranged?
"MISSISSIPPI" has 11 letters, with I repeated 4 times, S repeated 4 times, and P repeated 2 times.
Number of arrangements $= \frac{11!}{4! 4! 2!} = \frac{39916800}{1152} = 34650.$
- 14** In how many distinct ways can the letters of the word "STATISTICS" be arranged?
"STATISTICS" has 10 letters, with S repeated 3 times, T repeated 3 times, and I repeated 2 times.
Number of arrangements $= \frac{10!}{3! 3! 2!} = \frac{3628800}{72} = 50400.$
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Arrangements with restrictions

- 15** In how many ways can 5 people be seated in a row if two particular people must sit next to each other?
Treat the two people as one combined block: this gives 4 units to arrange, in $4! = 24$ ways. The two people within the block can be arranged in $2! = 2$ ways. Total: $24 \times 2 = 48$ ways.
- 16** In how many ways can 6 people be seated at a round table? (Rotations are considered identical arrangements.)
For circular arrangements, fix one person's position to remove rotational duplicates: $(6 - 1)! = 5! = 120$ ways.
- 17** In how many ways can 7 people be seated in a row if two particular people must NOT sit next to each other?
Total arrangements of 7 people: $7! = 5040$. Arrangements with the two together (treat as one block: 6! ways to arrange the 6 units, times $2!$ for the internal order): $6! \times 2! = 720 \times 2 = 1440$. Arrangements with them NOT together: $5040 - 1440 = 3600$.
- 18** In how many ways can 8 people be seated at a round table if two particular people must sit next to each other? (Rotations are considered identical arrangements.)
Treat the two people as one block: this leaves 7 units to arrange in a circle, giving $(7 - 1)! = 6! = 720$ ways. The two people within the block can be arranged in $2! = 2$ ways. Total: $720 \times 2 = 1440$ ways.
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Combining permutations and combinations

- 19** A team of 3 is chosen from 5 boys and 4 girls, and must include exactly 2 boys and 1 girl. In how many ways can this be done?
Choose 2 boys from 5: ${}^5C_2 = 10$. Choose 1 girl from 4: ${}^4C_1 = 4$. Total: $10 \times 4 = 40$ ways.
- 20** A committee of 4 is chosen from 6 men and 3 women, and must include at least 1 woman. In how many ways can this be done?
Total ways to choose 4 from all 9 people: ${}^9C_4 = 126$. Ways with no woman (all 4 from the 6 men): ${}^6C_4 = 15$. Ways with at least 1 woman: $126 - 15 = 111$.
- 21** A hand of 5 cards is dealt from a group of 4 aces and 8 other cards (12 cards total). Find the number of ways to choose a hand containing exactly 2 aces.
Choose 2 aces from 4: ${}^4C_2 = 6$. Choose 3 from the remaining 8 cards: ${}^8C_3 = 56$. Total: $6 \times 56 = 336$ ways.

Using combinations in probability

- 22** A bag contains 6 red and 4 blue balls. Three balls are drawn at random without replacement. Find the probability that all three are red.
 $P(\text{all red}) = \frac{{}^6C_3}{{}^{10}C_3} = \frac{20}{120} = \frac{1}{6}$.
- 23** A bag contains 5 red and 5 blue balls. Two balls are drawn at random without replacement. Find the probability that both are blue.
 $P(\text{both blue}) = \frac{{}^5C_2}{{}^{10}C_2} = \frac{10}{45} = \frac{2}{9}$.
- 24** A bag contains 7 red and 3 blue balls. Two balls are drawn at random without replacement. Find the probability that at least one is blue.
 $P(\text{no blue}) = \frac{{}^7C_2}{{}^{10}C_2} = \frac{21}{45} = \frac{7}{15}$. $P(\text{at least one blue}) = 1 - \frac{7}{15} = \frac{8}{15}$.

A well-designed word problem: forming a school quiz team

- 25** This question has two parts. A school has 12 eligible students (7 in Grade 11 and 5 in Grade 12) trying out for a quiz team of 4 students. (a) In how many ways can the team of 4 be chosen from all 12 students, with no restriction? (b) In how many ways can the team be chosen if it must contain exactly 2 students from each grade?
(a) ${}^{12}C_4 = \frac{12!}{4!8!} = \frac{12 \times 11 \times 10 \times 9}{24} = 495$. (b) Choose 2 from Grade 11: ${}^7C_2 = 21$. Choose 2 from Grade 12: ${}^5C_2 = 10$. Total: $21 \times 10 = 210$ ways.